ENTERPRISE CAMPUS ACADEMIC SYSTEM

Cognito Tech Hub

Autonomous Campus Technical Training, Verifiable Attendance & Placement Management Platform

RELEASE v2.4.0
DATE: September 2026
STATUS: STABLE • LIVE ON CAMPUS
DISTRIBUTION: OFFICIAL ENTERPRISE

Publisher: Cognito Insights Private Limited

Mobile Engine: Flutter 3 / Android Native

Cloud Core: Node.js TypeScript REST API

Database: Supabase PostgreSQL (Multi-Tenant)

09:00 – 16:00	0 Seconds	100% Isolated	74.4 MB
DAILY ATTENDANCE HOURS	ROLLING LOCKOUT (ELIMINATED)	INSTITUTIONAL DATA PRIVACY	PRODUCTION RELEASE APK

1. Executive Overview

Cognito Tech Hub is a purpose-built, enterprise-grade educational ecosystem engineered to modernize and digitize technical skill-building, corporate placement bootcamps, and classroom attendance tracking across autonomous and accredited engineering colleges.

Traditional educational management systems often suffer from fragmented manual paper rolls, vulnerable proxy attendance, cross-department data spillage, and inflexible 24-hour rolling locks that frustrate students who attended afternoon sessions and are blocked the following morning. Cognito Tech Hub solves these challenges through an end-to-end synchronized architecture combining a high-performance Flutter mobile application, role-restricted administrative dashboards, and a hardened backend API.

Institutional Value Proposition

- **Autonomous Campus Portals:** Dedicated, strictly isolated portals for College Admins and Training & Placement Officers (TPO).
- **Verifiable Presence:** Dynamic, cryptographically time-stamped QR codes clamped strictly to daily classroom hours.
- **Nominal Register Automation:** Instant department filtering (CSE, ECE, MECH, CIVIL, AIDS, IT) with one-click nominal export.
- **Holistic Analytics:** 360-degree student evaluation tracking technical attendance, punctuality, and learning behaviour.

Student & Trainer Empowerment

- **Zero-Friction Marking:** Fast camera-based QR scanning with instant database synchronization and visual confirmation.
- **Clean Daily Window:** Attendance opens freshly each day at 09:00 AM IST without any rolling 24-hour penalty.
- **Curated Study Handouts:** Instant offline-friendly access to syllabus materials, code notebooks, and lecture notes.
- **Digital Identity Verification:** Official institutional photo ID card with verifiable QR code and department metadata.

2. System Architecture & Multi-Tenant Isolation

The platform follows a modern, decoupled multi-tier architecture designed for horizontal scalability, zero-trust network transmission, and complete institutional data confidentiality.

HIGH-LEVEL TOPOLOGY END-TO-END SECURE FLOW

```
[ Flutter Mobile Client ] • Students • Trainers • Institutional Admins • TPOs
                                    • (HTTPS / TLS 1.3 / Cloudflare Zero Trust Tunnel)
[ Reverse Proxy & API Gateway ] • Rate Limiting • Tenant Resolution Middleware • CORS
                                    • (JWT Bearer Token with Embedded Institutional Claims)
[ Node.js / Express Core ] • Attendance Engine • Analytics • Roster • Materials
                                    • (Strict Tenant Scope: WHERE college_id = auth_college_id)
[ Supabase PostgreSQL 15 ] • Multi-Tenant Database • Row-Level Security • Temporal Indices
```

Technology Matrix

LAYER	TECHNOLOGY	VERSION	KEY ARCHITECTURAL FUNCTION
Mobile Application	Flutter / Dart	3.x / 3.29	Cross-platform native UI, QR scanner camera integration, offline-aware cache, adaptive responsive layouts (supporting compact 320px screens up to large tablets).
Backend API Core	Node.js / Express / TypeScript	20 LTS / 5.x	Type-safe REST API server, temporal window calculation, cryptographic QR token verification, JWT authentication, and automated audit logging.
Relational Database	PostgreSQL / Supabase	15.x	Multi-tenant schema, strict foreign-key integrity, indexed attendance timestamps with `Asia/Kolkata` time zone casting.
Edge & Tunneling	Cloudflare Zero Trust	Managed	Encrypted outbound tunnel protecting backend origins, DDoS mitigation, public HTTPS termination (`*.trycloudflare.com`), and low-latency asset delivery.

3. Institutional Confidentiality & Tenant Isolation Guarantees

In educational consortiums and multi-campus setups, cross-institutional data leakage is a critical compliance violation. Cognito Tech Hub enforces **Zero Data Bleed** between partner colleges:

1. Token-Derived Identity Resolution

College Admins and TPOs have their institutional `college_id` signed directly into their cryptographic JWT session. Any attempt by client requests to pass spoofed `?college_id=...` query parameters is immediately intercepted and overridden by the backend tenant middleware.

2. Roster & Nominal Roll Isolation

Adarsh Engineering College administrators can only query students with `college_id = 'ADARSH'` (e.g. Roll 21HN1A5401). External campus administrators can only view their own registered campus rosters. Multi-campus cross-listing is strictly prohibited at the query level.

3. Scoped Faculty & Trainer Allocation

Trainers assigned to an institution are scoped solely to that campus. Faculty allocations and evaluations for other universities remain completely inaccessible.

4. Campus Notice Broadcast Isolation

Notices posted by an institutional TPO are broadcast strictly to students enrolled in that specific institution. Institutional announcements never leak across campus boundaries.

4. The Daily Attendance Engine (09:00 AM – 04:00 PM IST)

The core innovation in the latest release is the strict daily working hours attendance window and the total elimination of rolling 24-hour lockouts.

CORE BUSINESS RULE SPECIFICATION STRICT ENFORCEMENT

- **Operating Window:** Attendance is strictly active between **09:00:00 AM** and **04:00:00 PM IST**.
- **Day-Scoped Calendar Reset:** Attendance uniqueness is strictly evaluated against the calendar date `(marked_at AT TIME ZONE 'Asia/Kolkata')::DATE = CURRENT_DATE`. No 24-hour rolling delay.
- **After 04:00 PM:** The attendance scanning bar is completely hidden; students receive an amber timeout notification.
- **Next Morning 09:00 AM:** The attendance bar reopens automatically for the new calendar day.

Dynamic 3-State Lifecycle

1. NOT_OPEN_YET (Before 09:00 AM)
Displays an indigo card: *"Attendance Opens at 09:00 AM. Scanner activates at 09:00 AM."* QR button is hidden.

2. OPEN (09:00 AM to 04:00 PM IST)
Active window. Unmarked students see **"SCAN ATTENDANCE QR"**. Marked students see **"TODAY'S ATTENDANCE COMPLETED"**.

3. TIMED_OUT (04:00 PM Onward)
The attendance bar and scan button are hidden. Displays amber card: *"Attendance bar reopens tomorrow morning at 09:00 AM."*

Backend Clamping Security

Even if a trainer or administrator generates a classroom QR slot late in the afternoon (e.g., at 03:30 PM with a 2-hour duration), the backend function `clampSlotExpiryTo4PM()` automatically clamps the slot expiration to **04:00:00 PM IST sharp**. No QR token remains valid after 4:00 PM.

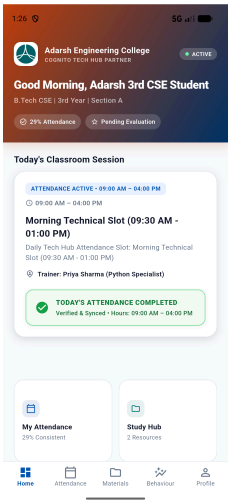


Figure 1: Live Student Dashboard during 09:00 AM – 04:00 PM IST window showing verified attendance completion.

Technical Implementation Details

COMPONENT	IMPLEMENTATION CODE	BEHAVIOR
`timeWindow.ts`	`isWithinAttendanceWindow(now)`	Evaluates current Indian Standard Time (UTC+05:30). Returns `true` strictly when hour ≥ 9 and hour < 16.
`attendance.routes.ts`	`POST /api/attendance/mark`	Rejects submissions outside 9AM-4PM with `400 Bad Request` and message *"Attendance is closed. Working hours are 09:00 AM to 04:00 PM."*
`student_dashboard_screen.dart`	`Timer.periodic(30s)`	Real-time 30-second timer re-evaluates the window. At 04:00:00 PM, the screen transitions automatically without requiring app restart.

5. Student Mobile App Experience

The student application is built for intuitive single-tap navigation, verifiable accountability, and immediate access to academic and placement resources.

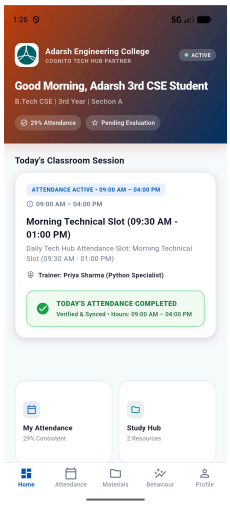


Fig 2.1: Classroom Session Card

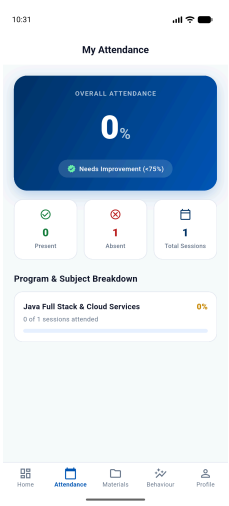


Fig 2.2: Attendance Analytics Tab

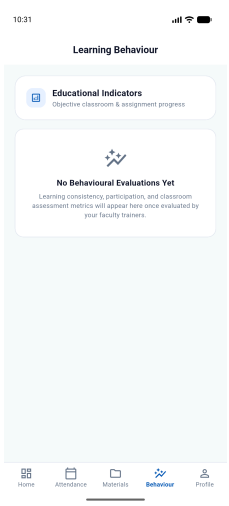


Fig 2.3: Behavioral Metrics Tab

Student Navigation Hubs

- **Home Hub (Tab 0):** Displays active classroom slot, assigned trainer details, today's attendance badge, and institutional affiliation notice.
- **Attendance Hub (Tab 1):** Complete month-wise attendance register, aggregate percentage score, presence consistency streak, and verified session logs.
- **Study Hub (Tab 2):** Digital repository of lecture handouts, PDF notes, coding reference sheets, and curriculum slide decks.
- **Behaviour Hub (Tab 3):** Real-time feedback from trainers scoring classroom punctuality, technical problem solving, teamwork, and engagement.
- **Profile Hub (Tab 4):** Official digital Student Identity Card with roll number, branch, academic year, and college affiliation emblem.

Security & Device Safeguards

- **Hardware Camera Scanner:** Embedded native camera barcode scanner utilizing ML-powered autofocus and edge detection.
- **Anti-Proxy Lock:** One mark per student per calendar day. Scans are cryptographically signed with the authenticated user ID and device timestamp.
- **Network Resiliency:** Automatic retry with exponential backoff on flaky campus Wi-Fi or low cellular coverage.
- **Responsive Ergonomics:** Tested and validated to render flawlessly on compact 320×640 screens without text wrapping or layout overflow.

6. Faculty Trainer Portal

Trainers and technical instructors use a dedicated, role-authorized interface to manage sessions, project dynamic QR tokens, review attendance in real time, and upload learning handouts.

Key Trainer Capabilities

- **Dynamic QR Token Generator:** Creates time-bounded attendance QR codes projected on classroom smart-screens or projector displays.
- **Operating Hours Templates:** Built-in standardized slots aligned with official hours:
  - Morning Technical Slot (09:00 AM – 01:00 PM)
  - Afternoon Hands-on Slot (01:00 PM – 04:00 PM)
  - Full Day Intensive Bootcamp (09:00 AM – 04:00 PM)
- **Manual Attendance Override:** Allows instructors to manually mark or excuse students experiencing hardware/camera issues.
- **End-of-Day (EOD) Reports:** Summary of topics taught, student turnout rate, and assignment milestones logged directly to college management.

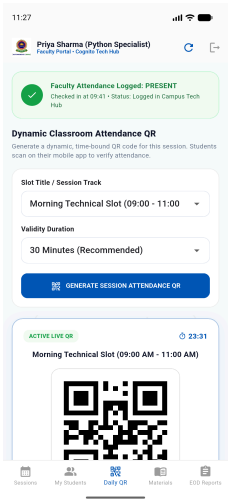


Figure 3: Trainer Dynamic Classroom QR Code Generator with real-time countdown timer.

7. Institutional TPO & College Admin Portal

The Training & Placement Officer (TPO) and College Admin interface provides confidential oversight of all enrolled students, assigned instructors, and placement readiness metrics.

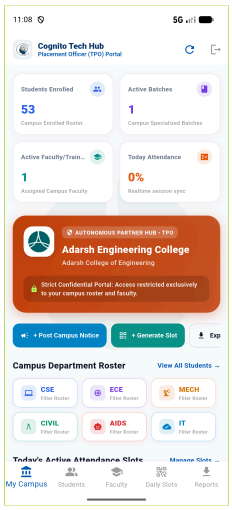


Figure 4: TPO Institutional Dashboard strictly isolated to campus roster and faculty.

Institutional Portal Modules

- **"My Campus" Overview (Tab 0):** Campus emblem, total student count enrolled at this institution, active campus batches, and assigned faculty roster.
- **Department Nominal Register (Tab 1):** Quick departmental filtering chips (`CSE` , `ECE` , `MECH` , `CIVIL` , `AIDS` , `IT`) with individual student attendance percentages and contact records.
- **Faculty Roster (Tab 2):** Institutional view of assigned technical trainers and their classroom evaluation metrics.
- **Campus Notice Board:** Instant broadcast tool to deliver important campus announcements and placement drive schedules.
- **Export Nominal Roll:** One-click export of complete student nominal registers to CSV or Excel for university compliance audits.

8. Technical & Build Specifications

PARAMETER	SPECIFICATION	NOTES
Application Name	Cognito Tech Hub	Official partner app for autonomous campus technical training.
Package Identifier	`com.cognitoinsights.techhub.cognito_techhub`	Unique Android Application ID.
Target Operating System	Android 5.0 (API 21) up to Android 15 (API 35)	Broad device compatibility across student smartphones.
Release Artifact	`app-release.apk` (74.4 MB)	Compiled with full optimization, icon tree-shaking (98.6% reduction), and release ProGuard rules.
Backend Port & Host	Port `5050` • `http://192.168.6.200:5050`	Node.js TypeScript runtime.
Cloud Tunnel	Cloudflare Zero Trust (Outbound HTTPS)	Encrypted edge proxy terminating at `*.trycloudflare.com`.

9. Production Download & Installation Links

The verified release APK and all associated backend services are live and available for immediate download and deployment:

DIRECT RELEASE APK DOWNLOAD (HTTPS CLOUDFLARE) RECOMMENDED • SECURE

https://placed-settlement-overview-virginia.trycloudflare.com/uploads/app-release.apk

Accessible from any Android mobile browser worldwide. Download and tap to install (enable "Install unknown apps" if prompted).

LOCAL CAMPUS LAN DOWNLOAD (ON-PREMISE) CAMPUS NETWORK

http://192.168.6.200:5050/uploads/app-release.apk

Direct high-speed LAN download when connected to the campus Wi-Fi network.

10. Installation & Setup Instructions

Step 1: Download

Open the download link in Chrome or any mobile browser on your Android device. The 74.4 MB APK will download directly.

Step 2: Install

Tap the downloaded file. When prompted by Android, grant permission to "Install Unknown Apps" for your browser.

Step 3: Login

Launch Cognito Tech Hub. Log in with your assigned institutional roll number/email and password to access your dashboard.

Cognito Tech Hub • Cognito Insights Private Limited
Confidential Enterprise Documentation • For Campus Distribution • All Rights Reserved © 2026